

3RD HARSH-ENVIRONMENT MASS SPECTROMETRY WORKSHOP

Ion Trap Secondary Ion Mass Spectrometry - Moving Toward Fieldable Systems

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ABSTRACT

In the event that chemical weapons are released somewhere in an urban area of the United States how will it be determined that the area is decontaminated to the point it can be opened to the public? Chemical weapons in an urban environment present unique challenges to decontamination and characterization because once released they are partitioned between different physical and chemical states, which dictate their mobility, toxicity, and susceptibility to treatment. CW's in the vapor state can be readily dispersed to non-toxic concentrations, are highly susceptible to environmental degradation, and the technologies for detecting them in the gas phase are relatively mature. However, adsorption on solid surfaces presents a difficult challenge for both detection and decontamination, because many CW's can bind tightly to a surface, resisting extraction or degradation, and can change form into equally toxic compounds while on the surface, confounding detection technologies tuned only to the agent compound. So, once the air has cleared, and the volatile components have dispersed, how is it determined if there remain on the surface, the steps, the handrails, the benches, a tightly bound concentration of agent, or equally toxic byproducts of the environmental or intentional degradation of the agent. Ion trap secondary ion mass spectrometry (ITSIMS) has been applied for detecting and characterizing trace levels of a variety of complex compounds on surfaces. In most cases the goal is simple detection and chemical identification. Past experience has demonstrated that levels down to 0.001 monolayer can be detected on many type of environmental samples such a soil, leaves, minerals, and concrete . In previous studies involving the detection of the nerve agent VX on different substrates it was noted that the degradation of the VX was highly dependent upon the substrate. For sandy soils degradation was extremely slow (months), while for concrete samples the degradation was extremely rapid (hours). For obvious reasons the lifetime of such compounds on different surfaces is an important factor in making operational decisions in the field, and thus there is impetus to develop field transportable systems. This presentation will summarize our current progress in development of field transportable ion trap SIMS.